WORKED SOLUTION

STEP II 2007, Question 7

Pure • Jensen's Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q7

This question has a full guided treatment in *Beyond STEP*, Section 3.3, *The Power of Convexity*: https://beyondstep.keithmcnulty.org/ch03/convexity.html

The key idea

Jensen's inequality says that for a concave function the average of the values is at most the value at the average. Each part comes down to spotting which concave function to use and which numbers to average. For the triangle, the angles have a fixed sum, so their average is known. For the AM–GM inequality, the logarithm turns a product into a sum. Parts (a) and (b) then need the right list of numbers to feed into AM–GM, including some constants.

Solution

Concavity

If $f(x) = \sin x$ then $f''(x) = -\sin x$. For $0 < x < \pi$ we have $\sin x > 0$, so $f''(x) < 0$ and $\sin x$ is concave there.

If $f(x) = \ln x$ then $f'(x) = 1/x$ and $f''(x) = -1/x^2$. This is negative for every $x > 0$, so $\ln x$ is concave for $x > 0$.

Part (i)

The angles of a triangle satisfy $A, B, C > 0$ and $A + B + C = \pi$, so each lies in $(0, \pi)$, where sin is concave. Jensen's inequality with $n = 3$ gives

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin\left(\frac{A + B + C}{3}\right) = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Multiplying by 3,

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2},$$

as required. Equality holds exactly when $A = B = C = \pi/3$, the equilateral triangle.

Part (ii)

Take $f(x) = \ln x$, which is concave on $x > 0$, and put $x_k = t_k > 0$. Jensen's inequality gives

$$\frac{1}{n}\sum_{k=1}^{n} \ln t_k \leq \ln\left(\frac{t_1 + t_2 + \cdots + t_n}{n}\right).$$

The left-hand side is $\frac{1}{n}\ln(t_1 t_2 \cdots t_n) = \ln \sqrt[n]{t_1 t_2 \cdots t_n}$. Since e^x is increasing, we may exponentiate both sides without changing the direction of the inequality:

$$\sqrt[n]{t_1 t_2 \cdots t_n} \leq \frac{t_1 + t_2 + \cdots + t_n}{n},$$

with equality if and only if $t_1 = t_2 = \cdots = t_n$. (For $n = 1$ this is trivially an equality, which is consistent.)

(a)

? **Ask yourself** Which four numbers should we average?

The left-hand side has four terms, so use $n = 4$. We need the geometric mean to produce xyz, and x^4, y^4, z^4 give $\sqrt[4]{x^4y^4z^4} = xyz$. The constant 16 is the fourth term, and $\sqrt[4]{16} = 2$. So take $t_1 = x^4$, $t_2 = y^4$, $t_3 = z^4$, $t_4 = 16$, all positive:

$$\frac{x^4 + y^4 + z^4 + 16}{4} \geq \sqrt[4]{16\,x^4y^4z^4} = 2xyz,$$

using $x, y, z > 0$ to take the fourth root. Multiplying by 4 gives

$$x^4 + y^4 + z^4 + 16 \geq 8xyz,$$

as required. Equality needs $x^4 = y^4 = z^4 = 16$, that is $x = y = z = 2$.

(b) Now the powers are fifth powers, so we want $n = 5$ with x^5, y^5, z^5 among the numbers. To get exactly xyz from the fifth root, the other two numbers must multiply to 1. The natural choice is $t_4 = t_5 = 1$:

$$\frac{x^5 + y^5 + z^5 + 1 + 1}{5} \geq \sqrt[5]{x^5y^5z^5 \cdot 1 \cdot 1} = xyz.$$

Rearranging,

$$x^5 + y^5 + z^5 - 5xyz \geq -2$$

for all positive x, y, z.

Ask yourself Does this show that the minimum is -2?

Not on its own: a lower bound is the minimum only if it is attained. Equality in AM–GM holds when all five numbers are equal, $x^5 = y^5 = z^5 = 1$, so $x = y = z = 1$. There the expression is $1 + 1 + 1 - 5 = -2$. Hence the minimum value is

$$\boxed{-2}, \quad \text{attained at } x = y = z = 1.$$

Checks

- In (i), a right-angled isosceles triangle gives $1 + 2 \cdot \frac{\sqrt{2}}{2} \approx 2.414$, below $3\sqrt{3}/2 \approx 2.598$. Random sampling of 200 000 triangles never exceeded 2.598.
- In (a), at $x = y = z = 2$ both sides equal 64.
- In (b), a numerical minimisation from random starting points always converged to $(1, 1, 1)$ with value -2. Also, $x = y = z = 1.1$ gives $3(1.61051) - 5(1.331) \approx -1.82 > -2$.

Common slips

- Forgetting that exponentiating preserves the inequality only because e^x is increasing; it deserves a word.
- In (b), stopping at "≥ -2" without exhibiting values of x, y, z where -2 is reached.
- Using AM–GM with $n = 3$ on x^5, y^5, z^5, which gives $x^5 + y^5 + z^5 \geq 3(xyz)^{5/3}$ and doesn't compare cleanly with $5xyz$.

Beyond STEP

Jensen's inequality is a statement about convex (or concave) functions, and its natural home is analysis and probability: in the language of expectation it reads $\mathrm{E}[f(X)] \leq f(\mathrm{E}[X])$ for concave f. The trick in (a) and (b), padding with constants so that the geometric mean comes out right, is the starting point of the theory of weighted AM–GM and Hölder's inequality. For more on inequalities and convexity, try Bartle and Sherbert, *Introduction to Real Analysis*; for the probabilistic version, Grimmett and Welsh, *Probability: An Introduction*.

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